

Terminalx shapes_exampl...
(github-pkg) codio@athenaLight-nevadaLopez:~/workspace/example3\$

Guide x

Collapse

Installing from GitHub

Installing Packages from GitHub

GitHub is one of the most popular code repositories. You can publish and share your projects with the world. This also includes the ability to upload share Python packages. Instead of using the traditional `git` tools to clone a repository, you can install the package directly from the GitHub website with `pip`.

Before we do this, let's create the `example3` directory and set up a virtual environment. We will name this environment `github-pkg` since we are installing a package from GitHub.

```
mkdir example3
cd example3
python3 -m venv github-pkg
source github-pkg/bin/activate
```

Head on over to the GitHub repository for the package. You can explore the different directories and folders to get an idea of what this package does.

▼ Did you notice?

There is no `requirements.txt` file because no third-party packages are used. There is also a `setup.py` file which is required for installing a package. All of the packages hosted on PyPI have this file as well. You do not need a `setup.py` file if you are writing your own packages and not hosting them elsewhere.

To install directly from GitHub, you are going to start your command with `python3 -m pip install` like normal. There are several ways of communicating with GitHub. We are going to use HTTPS, so the next part becomes `git+https://`. Next, add the URL from the address bar for the GitHub repository. The final thing you need to add is `#wheel=shapes`.

▼ What is a wheel?

Wheels are an important part of the package installation process. They allow for faster and more stable downloads and installations. We will talk more about wheels in the next assignment.

Put it all together, and this is the command to install the `shapes` package from its GitHub repository. Run the command in the terminal.

```
python3 -m pip install git+https://github.com/codio-content/shapes-package#wheel=sh
```

Testing the Shapes Package

To test the newly installed package, we first need a Python file in which we can write code. Use the `touch` command to create an empty file called `shapes_example.py`.

```
touch shapes_example.py
```

The link below will open the blank file. The name of the file **must** be `shapes_example.py`.

Open shapes_example.py

Copy/paste this script into the Python file. The `shapes` package has three classes, `Square`, `Circle`, and `Rectangle`. Create an instance of each class and call the `area()` method for each object.

```
from shapes.square import Square
from shapes.circle import Circle
from shapes.rectangle import Rectangle

s = Square(5)
c = Circle(10)
r = Rectangle(5, 10)

print(f"The square's area is {s.area()}")
print(f"The circle's area is {c.area()}")
print(f"The rectangle's area is {r.area()}")
```

Click on the tab for the terminal and enter the command below. Press `ENTER` to run the script.

```
python3 shapes_example.py
```

If everything works as intended, you should see the following output:

```
The square's area is 25
The circle's area is 314.1592653589793
The rectangle's area is 50
```

Reading Question

True or False: The only place from which you can install packages is PyPI.

☐ True

☒ False

False. You can also install Python packages from other websites like GitHub. You can also download packages from other version control systems.

Check It!

Next